

Global Renewable Energy in 2025: How Far Have We Come?

A data-driven overview of renewable energy's growing share in the global energy and electricity mix, spanning hydropower, wind, and solar technologies.

For: Policy Analysts & Energy Sector Strategists

Source: Our World in Data – Renewable Energy | Ember (2026); Energy Institute – Statistical Review of World Energy (2025)

Solar

Wind

Hydro

Other

75% of Global GHG Emissions Still Come from Fossil Fuels

75%

of global greenhouse
gas emissions from
fossil fuel combustion

5M+

premature deaths per
year from fossil fuel
air pollution

~150

of fossil fuel
dominance since the
Industrial Revolution
years

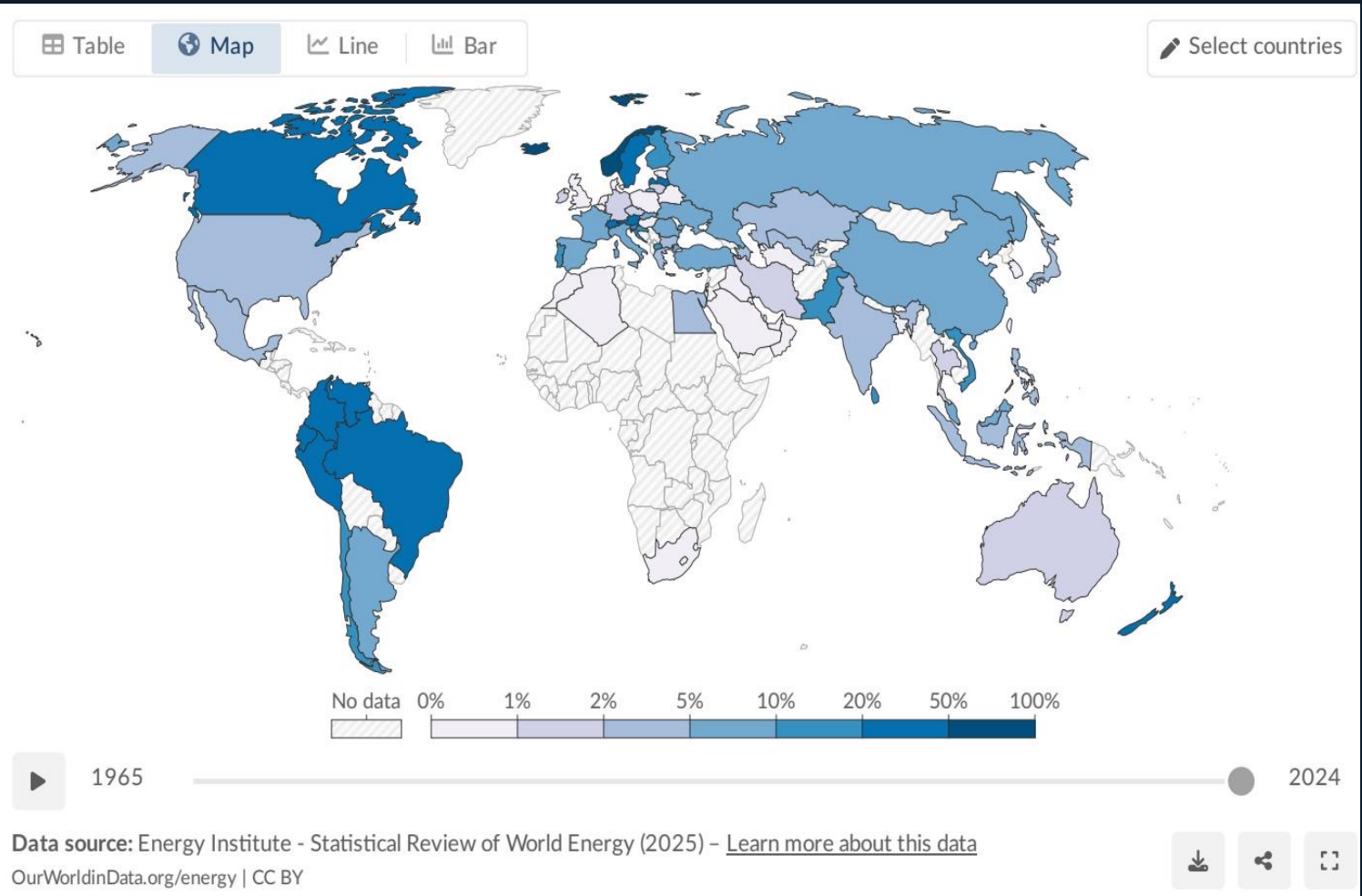
Since the Industrial Revolution, most countries' energy systems have become dominated by fossil fuels.

Fossil fuels drive both the climate crisis and a severe public health emergency through local air pollution.

To reduce CO₂ emissions and air pollution, the world must rapidly shift toward low-carbon energy sources — nuclear and renewable technologies.

⚠ The urgency is clear: transitioning to renewables is not just an environmental goal — it is a public health imperative.

Renewables Now Supply ~14% of Global Primary Energy



Measured using the substitution method

~14%

of global primary energy
now from renewables (2024)

Primary energy = electricity
+ transport + heating

Why lower than electricity?

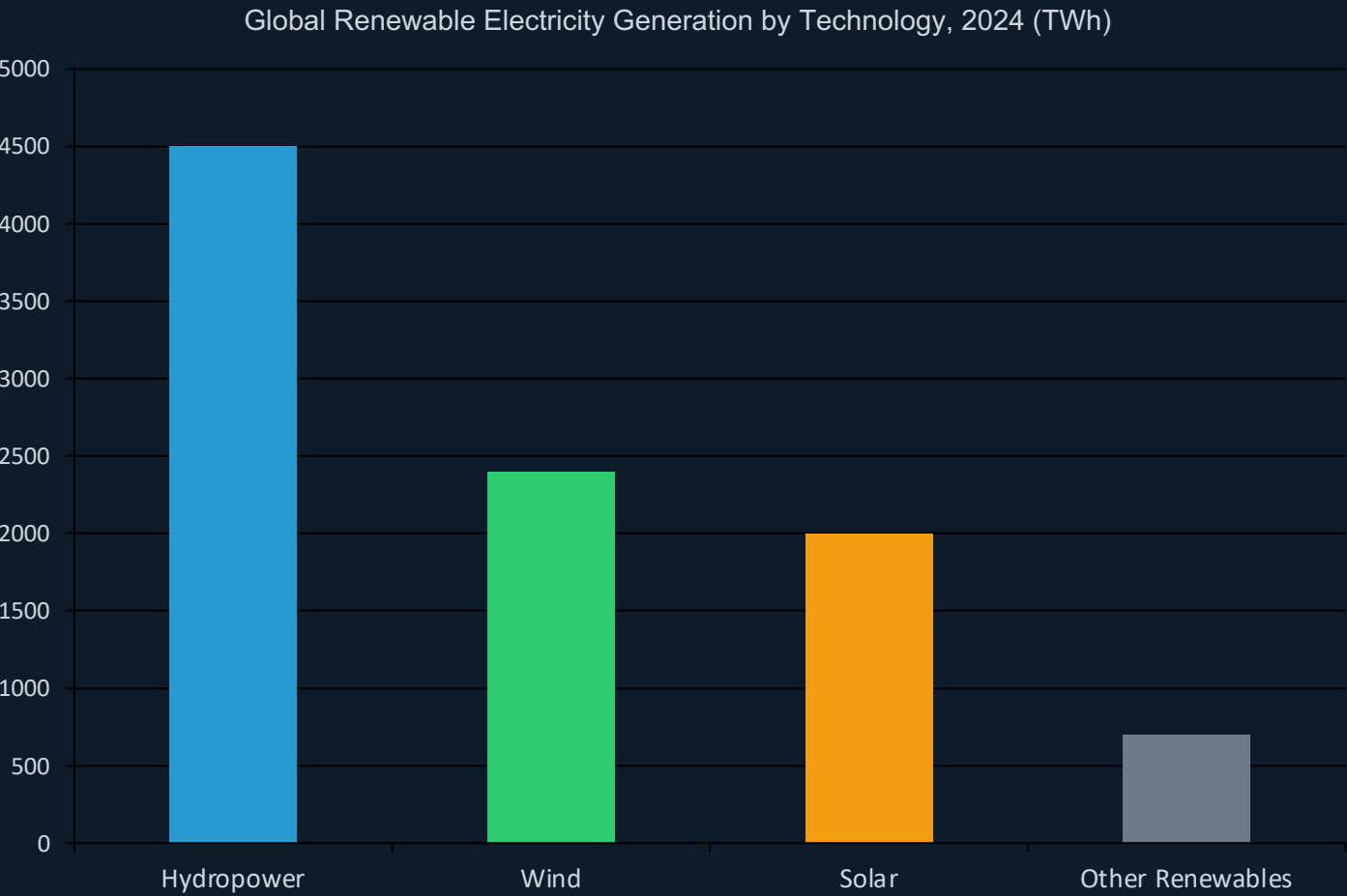
Transport and heating are
harder to decarbonize —
still reliant on oil and gas.

Regional leaders:

Norway, Brazil, New Zealand
derive large renewable shares
of primary energy.

Laggards: Middle East and
parts of Asia remain below 5%.

Renewable Electricity Generation Is Growing Rapidly, Led by Wind and Solar



Technology Breakdown (2024)

Technology	TWh	Share
Hydropower	~4,500	~47%
Wind	~2,400	~25%
Solar	~2,000	~21%
Other	~700	~7%
TOTAL	~9,600	100%

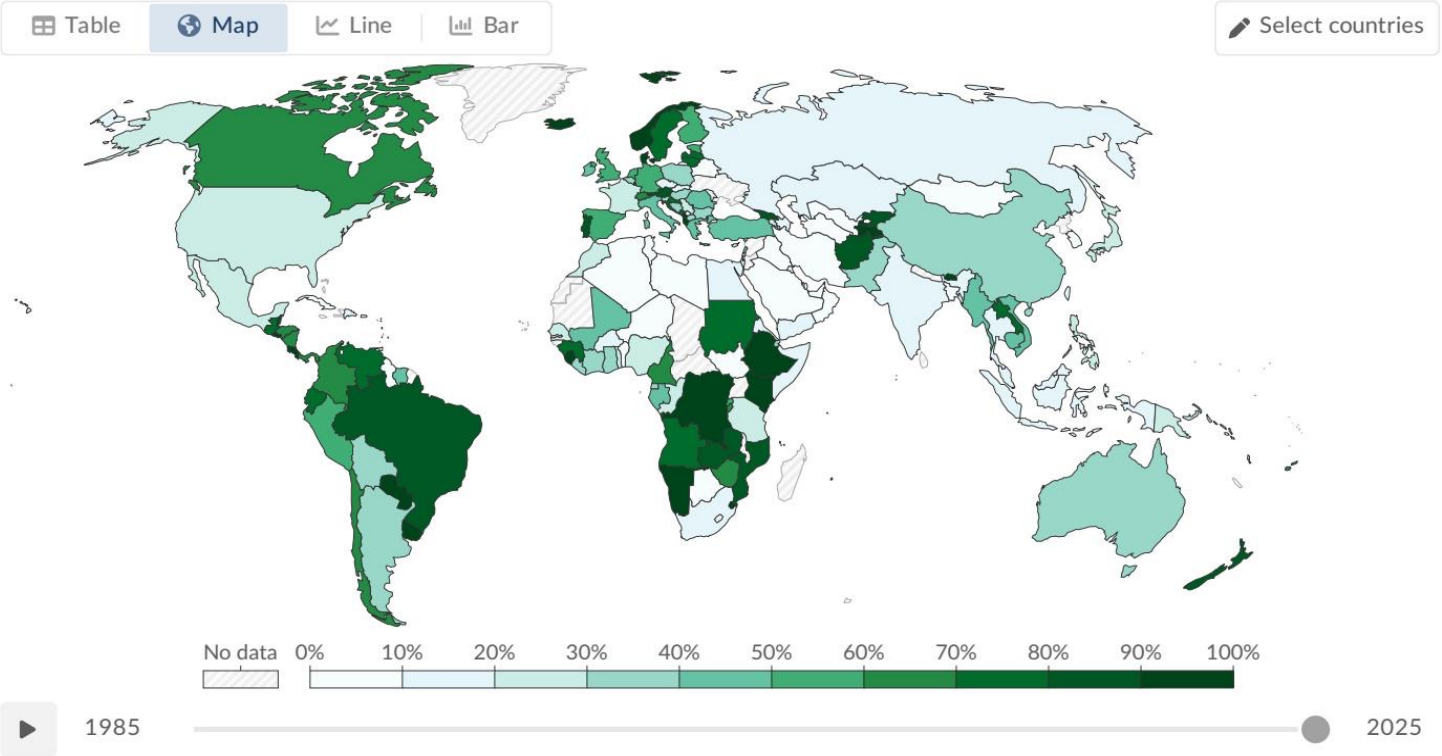
Wind + Solar combined (~4,400 TWh) now nearly match hydropower — and are growing far faster.

Almost One-Third of Global Electricity Now Comes from Renewables

Share of electricity production from renewables, 2025

Renewables include solar, wind, hydropower, bioenergy, geothermal, wave, and tidal sources.

Our World
in Data



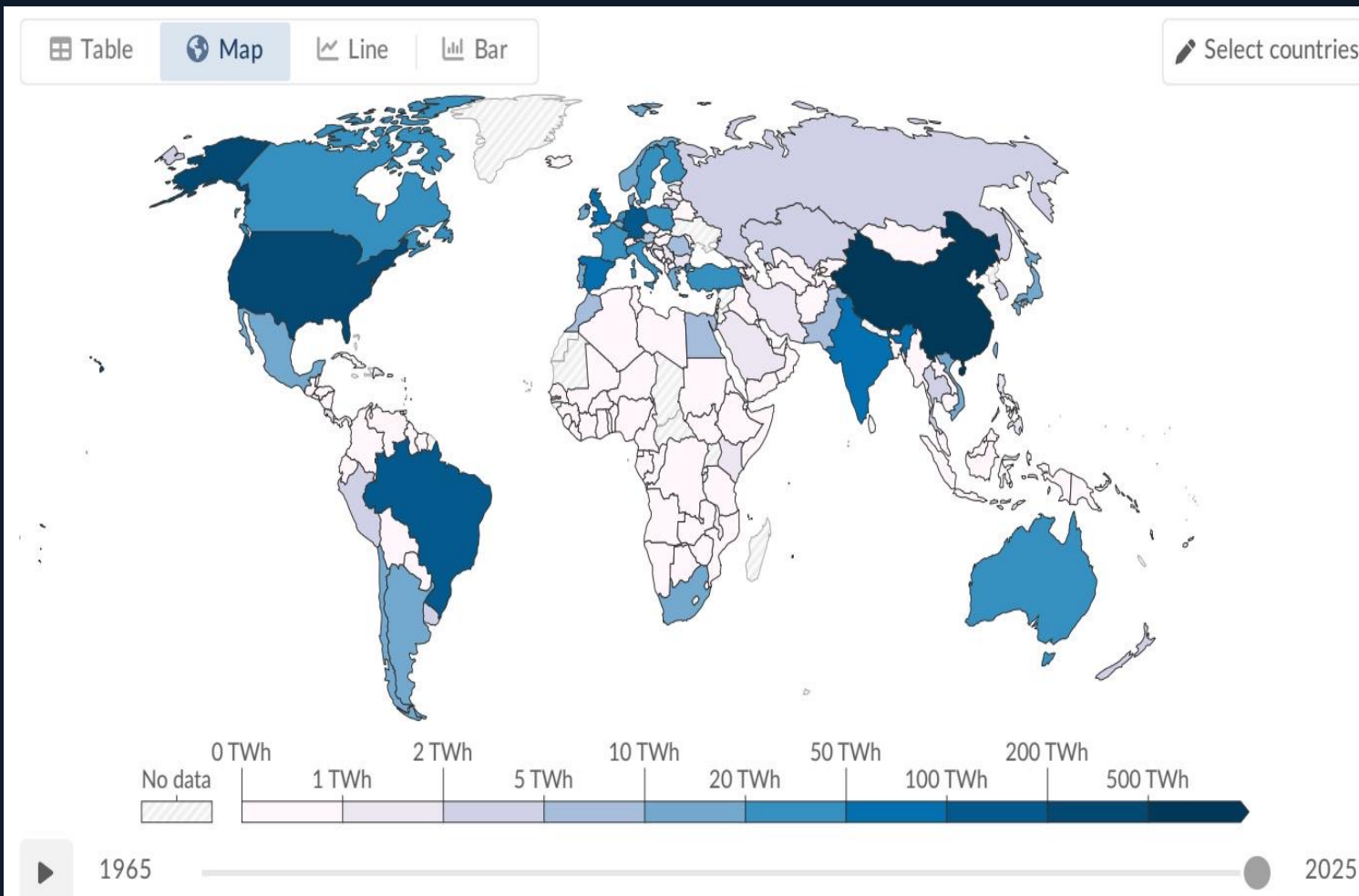
Data source: Ember (2026); Energy Institute - Statistical Review of World Energy (2025) – [Learn more about this data](#)
OurWorldinData.org/energy | CC BY

The electricity sector leads the transition: renewable share (~30%) is more than double the primary energy share (~14%), as electricity is easiest to decarbonize.

Selected Countries: Renewable Electricity Share (2025)

Country / Region	Share
Norway	~98%
Brazil	~85%
Canada	~65%
Germany	~55%
China	~35%
World avg.	~30%
India	~25%
United States	~25%
South Africa	~10%
Saudi Arabia	<2%

Hydropower Remains the Largest Renewable Source at ~50% of Generation



~4,500 TWh

global hydropower generation
(2024) — ~47% of all
renewable electricity

Top producers:

🇨🇳 China · 🇧🇷 Brazil · 🇨🇦 Canada

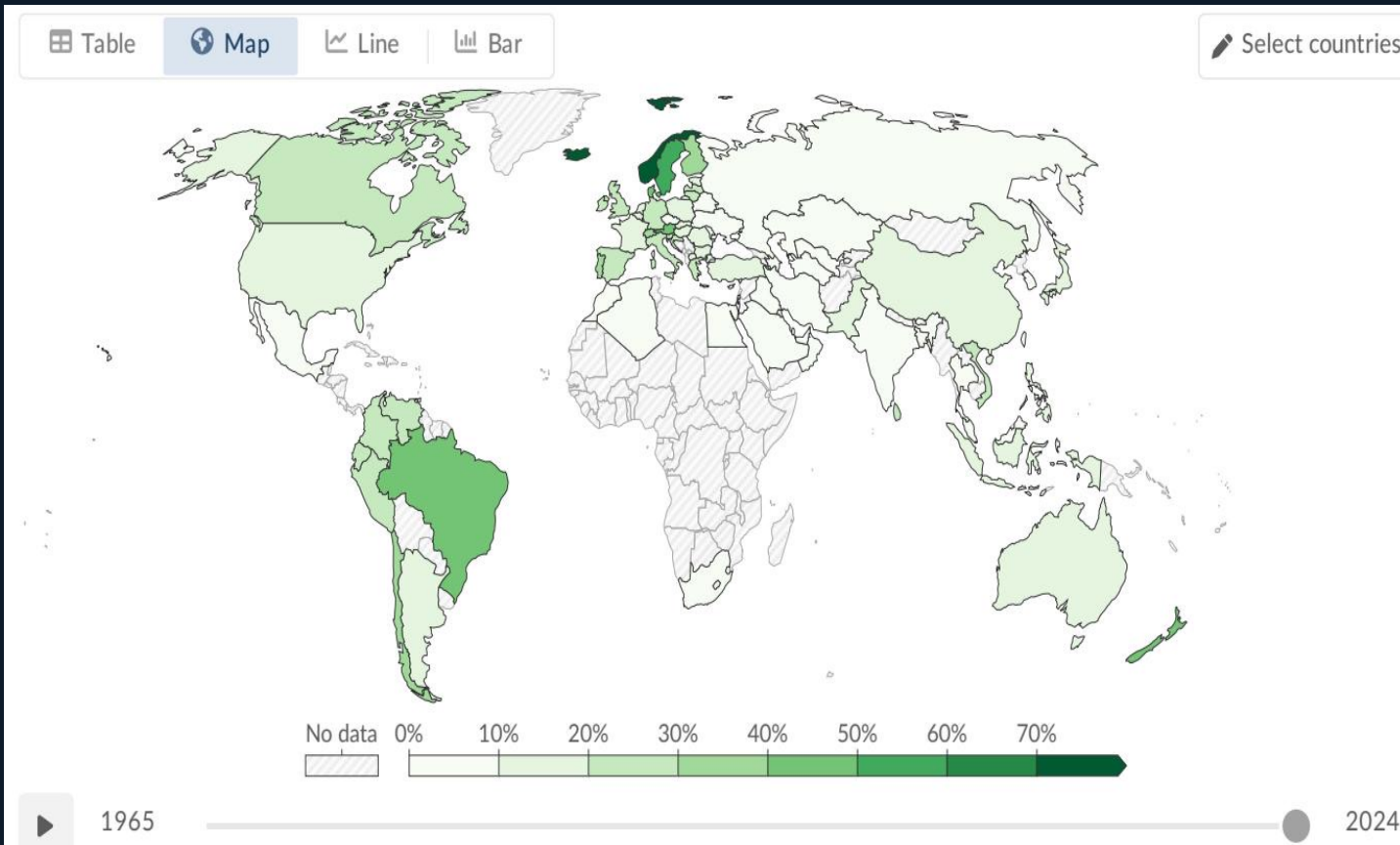
A century of scale:

Hydropower has generated
electricity at scale for over
100 years — the backbone of
renewable generation.

Growth outlook:

Hydro's expansion potential
is limited compared to wind
and solar.

Wind and Solar Are the Fastest-Growing Energy Technologies in History



Solar's rapid rise driven by plummeting costs and supportive policy since 2010.

WIND

~2,400 TWh

~25% of global renewable electricity (2024)
Onshore & offshore · leads in Europe, China, US

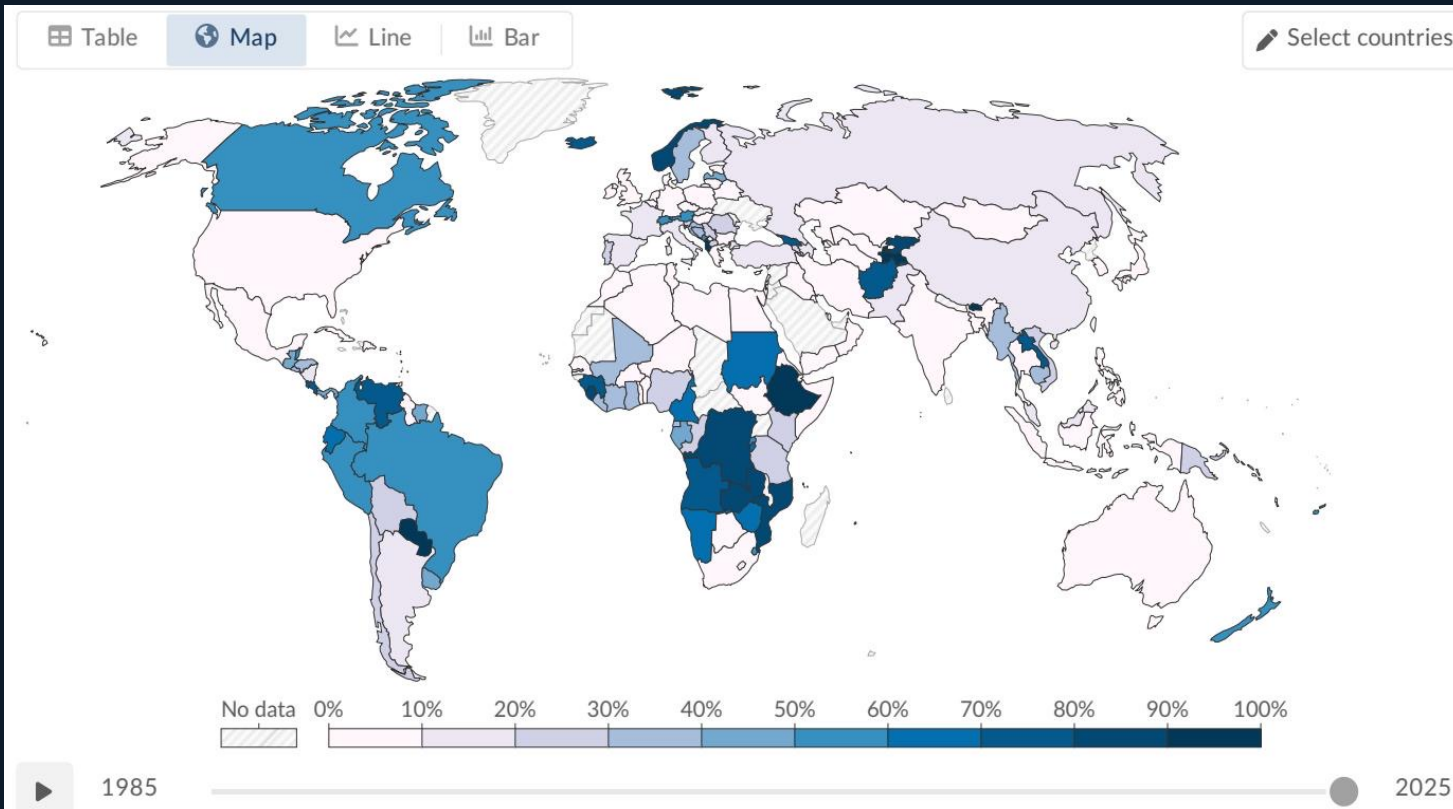
SOLAR

~2,000 TWh

~21% of global renewable electricity (2024)
Fastest-growing source · dramatic rise since 2010

Wind + Solar combined: ~4,400 TWh (~46% of renewable electricity)
— now nearly matching hydropower's ~4,500 TWh

Stark Regional Gaps: Sub-Saharan Africa Lags While Scandinavia and Latin America Lead



Map above: share of electricity from hydropower by country (2025)

Leaders

Norway ~98%
Brazil ~85%
Canada ~65%
Germany ~55%
Denmark >60%

Laggards

Saudi Arabia <2%
S. Africa ~10%
Sub-Saharan avg. <10%
Middle East avg. <5%

Why the gap?

- Scandinavian countries & Brazil: abundant hydropower resources
- Denmark & Germany: heavy investment in wind & solar
- Sub-Saharan Africa: minimal renewable infrastructure, financing gaps
- Middle East: almost entirely fossil-fuel dependent for electricity

Key Takeaways: Renewables Are Growing Fast but the Transition Must Accelerate

01

Significant milestone, but far from sufficient

~14% of global primary energy and ~30% of global electricity now come from renewables — a historic achievement, but well short of climate targets.

02

Hydropower still dominant, but momentum is shifting

Hydro generates ~half of all renewable electricity (~4,500 TWh), but its growth potential is limited. Wind and solar are catching up fast.

03

Wind and solar are the fastest-growing energy technologies in history

Combined (~4,400 TWh / ~46%), they now nearly match hydropower. Solar's dramatic cost decline since 2010 has been the key driver.

04

Stark regional disparities persist

Scandinavia and Latin America exceed 60–90%+ renewable electricity; much of the Middle East and Sub-Saharan Africa remain below 10%.

05

Transport and heating lag far behind electricity

Harder-to-decarbonize sectors keep the overall primary energy share much lower. Electrification of transport and heat is the next frontier.

The pace of the energy transition must accelerate. Scaling wind and solar, electrifying transport and heating, and closing the regional access gap are the defining energy policy challenges of this decade.